

22 Sep Assignment notes

Background

- STPs treat sewage, treated water released into lakes
- city is considering re-use instead of release
 - S1 ↳ irrigating farms on city's edge
 - S2 ↳ watering public parks
 - S3 ↳ cooling water to industrial estate
- Assumption: Re-use of water improves public health further

Research questions

- ① Does re-use improve public health?
- ② If it does, how valuable is the improvement?
- ③ How do public health benefits vary across stakeholder groups?
- ④ How sensitive are the values to assumptions?

Study Area

- ↳ Any Indian city
- ↳ if data borrowed from outside India, contextualise it.

Deliverables

- ↳ suggested 7pg pdf or 10 slides
- ↳ excel sheet w/ calculations
- ↳ explicit note on use of AI

My thoughts

- ↳ must conceptualise public health. I think they mean more of human physical health, less ecosystem / wildlife health. Does improvement in agri yield count as health? Would mental health count?

Look at CICES, literature on public health benefits from water quality improvements.

- ↳ choice of study area is key. If there is no demand or use of 'health' benefits, there's no value in improvement. For example, is lake water used for drinking? for bathing? This choice will affect every answer to RA + data availability

Look for Indian city, where benefits would make sense and would have data. Sai Krishna had suggested the open data platform.

- ↳ Steps by order → Past literature: Green book, consensus

- conceptualise Public health
 - causal diagram
 - study area choice
↓
 - biophysical impact data
 - monetary data / method
- if difficult change

Entity	Definitions/Descriptions	Citation	Suggested definition
Public Health	The art and science of health promotion and protection, disease prevention, and the improvement and prolonging of quality of life through the organized efforts of society.	(32–39)	Public health is an art and science, based on objective findings but responsive to the needs and contexts of populations, concerned with addressing the health needs of a community. It is a diverse set of organized activities aimed at improving quality of life and reducing health disparities to enable people to thrive.
	The organized efforts of society to prevent morbidity and premature mortality, keep people healthy, improve health and well-being, and reduce health inequalities.	(66)	
	The proactive approach to protecting the health of a community.	(93)	
	Public health fulfills society's collective interest in assuring environments that allow people to thrive.	(72)	
	The political art of applying science with the aim of reducing health inequalities while ensuring the overall health of the population.	(58)	
	Social medicine is the impact of decisions or policies made by other sectors that impact health, i.e. welfare, education	(79)	
	The diverse set of activities that focus on the promotion and protection of the health of the population and address health needs.	(77)	

CSE Report on wastewater reuse India 2024 Dec

Questions:

1. What are class 1 and class 2 cities
2. Difference between CETPs, TTPs, FSTPs and STPs

· Bangalore: 72% treated water used for lake revival
 Pg 137 21% " " " industrial & irrigation

Table
30



- seems like direct use is prohibited : domestic, drinking X
- but aquifer recharge → bore well → irrigation, drinking, domestic ✓
- closest to direct reuse @ Sihineeru Kere (Devanahalli)

From
 Hebbal
 Nagarava
 valley project



Source: BIOME Environmental Trust

Water treatment plant
 water quality monitored
 Annex 4

→ similar planned
 @ Doddakere lake

→ Raja Kaluve STP & Yelahanka STP
 sell treated water to industries @ (independent)
 (end-to-end) Rs 25/KL OR Rs 18/KL

Kutranjan et al 2024 Jan
 (WELL Labs) → How water flows through BIR

Pg 47

Table 9

estimated → amount of wastewater generated → 1940 MLD
 → treatment capacity in city (central & decentral) CSTP: 1239
 DSTP: 250
 → what happens to un/treated water CESTP: 5
 untreated: 218,
 266
 Reuse
 on spot: 28+19

Pg 50

Table 10

1040 into lakes (treated + untreated)
 472 into rivers

Scenarios

Green = WELL Labs data white = CSE data

BAU : Waste water from Bengaluru (1940 MLD) is treated by
 Scenario 1239 MLD mult. STPs. 825 MLD of treated water is being
 used/redirected for → lake recharge 72% - 1040 MLD
 → irrigation & industrial 21% (treated + untreated)
 → others (drains, indus, horti, etc) 7%

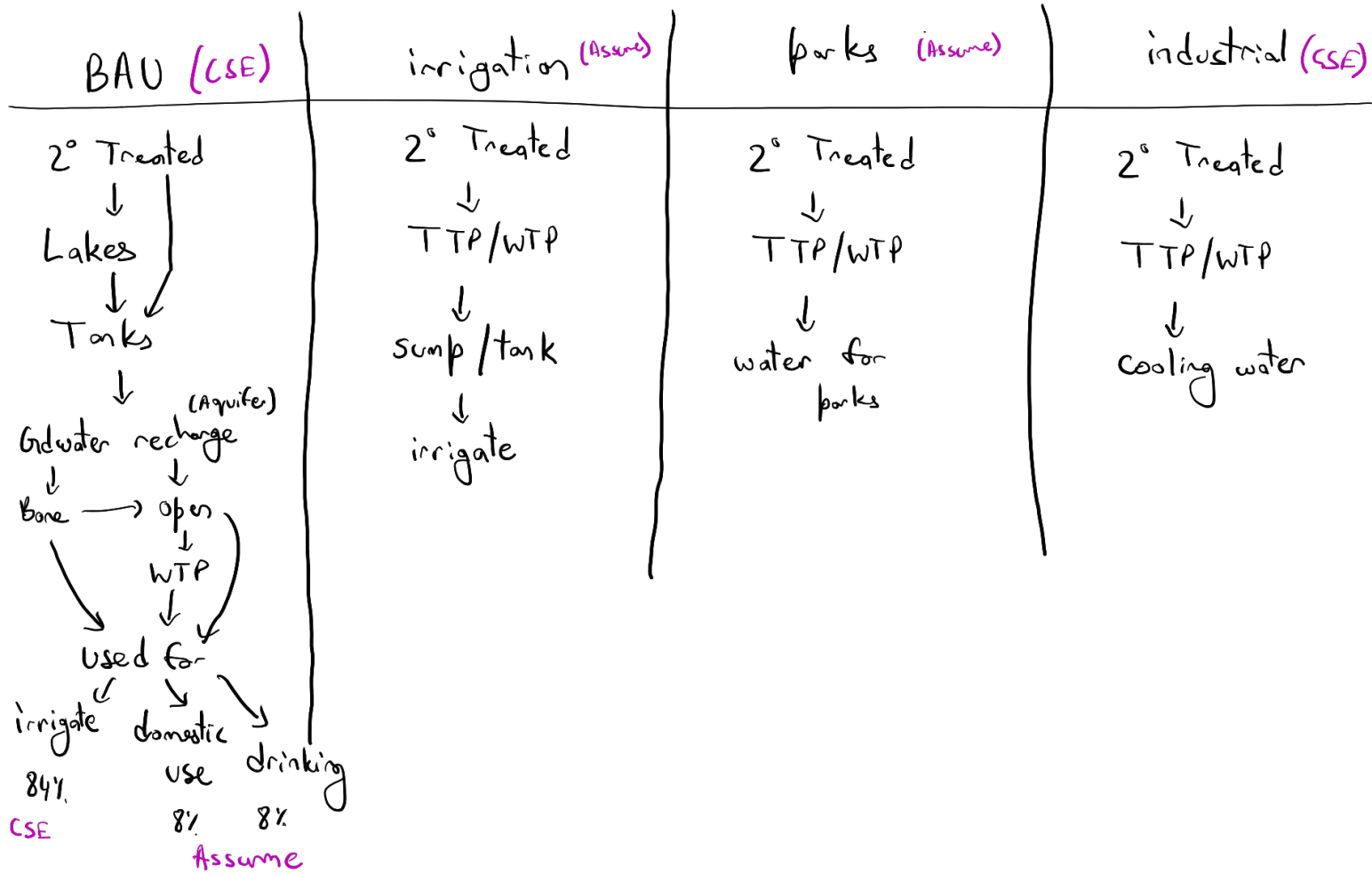
Reuse : More of the treated water used for :
 Scenarios

S1 ↳ irrigating farms on city's edge

S2 ↳ watering public parks

S3 ↳ cooling water to industrial estate

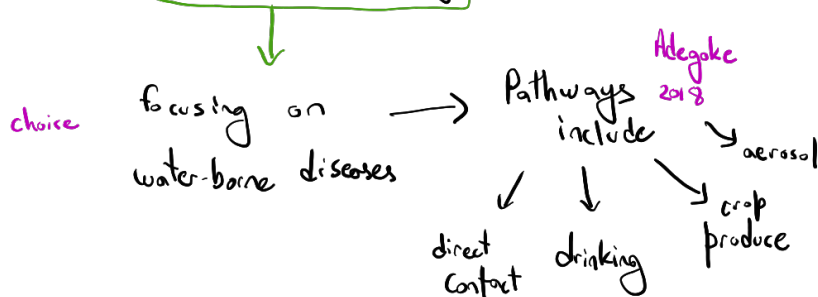
Water flow in each scenario



Effects on Public health

- Public health → Prevent / Reduce disease, premature mortality + Promote health, improve well being quality of life + Reduce health inequalities in a community

Jarvis 2020



- People / Groups who could be exposed to a type of contaminated water / product logical reasoning

BAU	irrigation	parks	industrial (cooling water)
Recreational users lake	Farm workers Tank or TTP water	Park visitors Aerosol	maybe workers TTP water
Farm workers & residents Bore or WTP water	crop consumers produce	& workers TTP water	

Rethinking RQs

Maybe it makes more sense to divide RQs by pathway. (exposure)

- ① What is the pathway of water in each scenario? ^{treated}
- ② Who is exposed to the TW ... ? ^{what is health risk posed by each scenario}
- ③ What is the conc. of pathogens in the water at point of exposure?
- ④ What is the COI / DALY per year due to exposure?
- ⑤ How sensitive is the value of COI / DALY to variables?

Mara et al. 2007

↳ estimated human health risk associated with using waste water for irrigation

QRA:

- ① id. hazards / pathogens that matter
- ② how much are ppl. exposed (dose)
- ③ dose-response: gives $P(\text{infection})$ & $P(\text{disease})$
- ④ estimate the population level risk

Kalita et al 2026

↳ Burden of diarrhoeal disease across India 1990-2023

Table 5
Number, age specific death and DALYs rate per 100,000 of Diarrhoeal disease, in 2023 (95%UI = 95% uncertainty interval.) by etiology.

Etiology	Deaths					DALYs				
	Number		Rates per 100,000 (95% UI)			Number		Rates per 100,000 (95% UI)		
	<5 Years	All Age groups	<5 years	Crude rate (All age group)	Age standardised rate (all age group)	<5 Years	All Age groups	<5 years	Crude rate (All age group)	Age standardised rate (all age group)
Adenovirus	12828 (6958-20582)	21187 (13894-30954)	11.7 (6.3-18.7)	1.5 (1.2-2.1)	2.1 (1.4-3)	1187079 (645925-1899854)	1538512 (966900-2261240)	1078.8 (587-1726.6)	106.6 (67-156.7)	141.4 (88.5-210.4)
Aeromonas	906 (357-1932)	2031 (1051-3564)	0.8 (0.3-1.8)	0.1 (0.1-0.2)	0.2 (0.1-0.3)	83901 (35172-178558)	132649 (67977-221228)	76.2 (32-162.3)	9.2 (4.7-15.3)	11.7 (6-20.4)
Campylobacter	3603 (1466-7453)	18927 (5963-38718)	3.3 (1.3-6.8)	1.3 (0.4-2.7)	2.2 (0.6-4.1)	340509 (145184-699069)	757590 (373210-1381093)	309.5 (131.9-635.3)	52.5 (25.9-95.7)	67.8 (34.3-125.8)
Cholera	743 (408-1225)	7125 (3822-12522)	0.7 (0.4-1.1)	0.5 (0.3-0.9)	0.6 (0.3-1)	65946 (36206-108794)	343244 (178852-568452)	59.9 (32.9-98.9)	23.8 (12.4-39.4)	25.7 (13.5-42)
Clostridium difficile	9 (4-16)	544 (324-964)	0 (0-0)	0 (0-0.1)	0 (0-0.1)	755 (376-1395)	16445 (9760-32332)	0.7 (0.3-1.3)	1.1 (0.7-2.2)	1.3 (0.8-2.4)
Cryptosporidium	3536 (1805-6220)	10925 (6917-16612)	3.2 (1.6-5.7)	0.8 (0.5-1.2)	1.2 (0.7-1.8)	323366 (166407-568543)	513056 (305955-769737)	293.9 (151.2-516.7)	35.5 (21.2-53.3)	47.4 (27.6-72.9)
Entamoeba	2973 (1177-6255)	14062 (6957-25592)	2.7 (1.1-5.7)	1 (0.5-1.8)	1.4 (0.7-2.6)	271565 (109984-561549)	587183 (306072-1014747)	246.8 (100-510.3)	40.7 (21.2-70.3)	51.5 (26.9-87.2)
Enteropathogenic E coli	4166 (1878-7475)	4620 (2284-8116)	3.8 (1.7-6.8)	0.3 (0.2-0.6)	0.4 (0.2-0.8)	383898 (173525-690559)	420562 (207265-735705)	348.9 (157.7-627.6)	29.1 (14.4-51)	39.6 (19.1-69.4)
Enterotoxigenic E coli	3745 (2107-6251)	13948 (8431-23032)	3.4 (1.9-5.7)	1 (0.6-1.6)	1.6 (0.9-2.6)	350527 (201869-581616)	596982 (395458-865901)	318.6 (183.5-528.6)	41.4 (27.4-60)	55.9 (36.5-79.9)
Non-typhoidal Salmonella	1005 (208-2229)	7236 (208-20425)	0.9 (0.2-2)	0.5 (0.1-4)	0.7 (0.2-1)	92302 (21723-204541)	243071 (37566-556432)	83.9 (19.7-185.9)	16.8 (2.6-38.6)	21.5 (3.2-49.7)
Norovirus	2362 (845-4600)	51715 (8826-97328)	2.1 (0.8-4.2)	3.6 (0.6-6.7)	5.2 (0.9-9.8)	228021 (91418-429649)	1508101 (471563-2697543)	207.2 (83.1-390.5)	104.5 (32.7-186.9)	126.9 (39.9-226.3)
Rotavirus	9387 (6251-14190)	25888 (18831-37448)	8.5 (5.7-12.9)	1.8 (1.3-2.6)	2.6 (1.9-3.8)	862151 (579382-1293109)	1425180 (1051843-1943554)	783.5 (526.5-1175.2)	98.7 (72.9-134.7)	128.6 (92.7-177.6)
Shigella	6069 (3288-10433)	13481 (8927-20564)	5.5 (3.9-5)	0.9 (0.6-1.4)	1.3 (0.8-1.9)	565797 (307361-965883)	942571 (624827-1392247)	514.2 (279.3-877.8)	65.3 (43.3-96.5)	82.2 (54-126)

DALY: disability-adjusted life-year.

Ramesh et al 2020 E coli Salmonella Crypto Giardia

↳ measured risk of infection from waste water reuse in agri, garden irrigation & industrial reuse

Pathogenic E. coli spp. is detected in all (100%) samples with a mean of 177.5 CFU/mL. Salmonella spp. is detected at 63% of sampling points with a mean of 4.7 CFU/mL. The findings were similar to previous study reports analysing pathogens in treated municipal wastewater (Kadam et al. 2008; Tyagi et al. 2008; Al-Gheethi et al. 2013).

The Cryptosporidium was detected in 81% (mean concentration of 3.8 oocysts/L) and Giardia in 45% (mean concentration of 1.9 cysts/L) samples.

↳ But seems like ETPs are only primary treatment plants

↳ Also estimated exposure scenarios based on surveys with users & ETP managers Table 3

↓
can use for
BAU + intervention
scenarios

Adegoke et al 2018

Many organisms

↳ has concentrations of organisms per waste water type (Treated vs. untreated)
↳ from various countries

George et al 2015

Ecdi Rota virus campylobacter

→ can use for BAU

↳ has concentrations of organisms in drinking water from Mysore

↳ Table 2: List of equations for disease burden calcs.

Bergion et al 2018

↳ has methods to value health risk reduction

various methods: WTP, CoI, value of DALY/QALY
(stated pref.)

↳ value of DALY $\rightarrow$ Human capital $\Rightarrow$ GDP/income per capita, Minimum wages per-yr
 $\rightarrow$ Friction cost method $\Rightarrow$ not sure

↳ cost of illness $\rightarrow$ direct healthcare costs
 $\rightarrow$ indirect cost (seems similar to DALY)

WHO 2001

↳ suggests DALY value at 3x income ^{national} per capita
but studies usually take 1x " "

Excel / Script

• What I need

↳ Exposure to organisms (dose) $\rightarrow$ how many ppl. are exposed to how much water (L) with how many organisms (log/L) } CSE Report
↳ Dose-response model
↳ $P(\text{disease} | \text{infection})$ George 2015 or Ramesh 2020 or Adegoke 2018
↳ disability weights
↳ value of a DALY

• Exposure (BAU)

(Devanahalli case study)

CSE Report has total water demand, %age satisfied by pumped water
 $\downarrow$
gives how much of the total water is treated water.
 $\downarrow$ STPs, $\downarrow$ Lakes, WTP

Say TT water = 100L $\rightarrow$ 84L goes to irrigate
 $\rightarrow$ 16L goes to domestic/drinking } Karnataka avg.

only population involved in agri. could be exposed to that 84L
and full population could be exposed to domestic / drinking 16L

Devanahalli → 38000 ppl. → all exposed to 16L of TT water

↳ agri workers exposed to 84L of TT water
(% of 38000)

Lakes near Devanahalli receive

water → 38000 ppl.

500 MLD

water → × ppl.

↓
Total num
of
ppl. exposed

↓
full on
part of
it

• Exposure (irrigation)

↳ Treated water directly goes to sump. ←
↓
only agri workers exposed

↳ Assume same population / % of tot. pop based on amount of TW diversion

• in both cases, exposure rate

↳ Drinking: 100% water ingested

↳ irrig/domestic: a % of water could be ingested

Sensitivity Analysis

↳ just choose highly uncertain variants according to what you noticed during analysis.